Signal Propagation Over Polarization-Maintaining Fibers: Problem and Solutions

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***Abstract*—Polarization-maintaining (PM) fibers are able to preserve the state of polarization (SOP) of a signal in the fiber reference frame. The SOP follows one of the axes of the fiber defined by the mechanical constraint that has been inserted during manufacturing. This ability is characterized by the polarization extinction ratio, which should be as high as possible. It is known to be very high (often above 50 dB) for the fiber itself. However, the somewhat low polarization extinction ratios (often between 20 and 30 dB) of connections (connectors, splices, . . .) induce strong distortions of the signal. The authors first explain why for one PM patch cord, i.e., a fiber and its connectors, with numerical and analytical models. They then extend these models to a patch-cord concatenation. Finally, they show and compare analytically and numerically different solutions to this problem: fiber length reduction, polarizer insertion, high-performing connectors, and axis alternation. The latest solution consisting in alternating the slow and the fast axes from one fiber to the next one is particularly efficient. They have implemented these solutions for the purpose of the source of the "Laser MégaJoule," and it exhibits far better performance.**

***Index Terms*—Distortion, laser fusion, optical communication, optical fiber polarization.**

I. Introduction

PROPAGATION over polarization-maintaining fibers (PMFs) is an old topic of fiber optics [1], [2]. PMFs are used to preserve the state of polarization (SOP) of one signal in the fiber reference frame that is defined by the mechanical constraint that has been added during the manufacturing process. PMFs allow to make two signals interact along a fiber avoiding fluctuations of their polarization states or to use polarization-dependent devices such as electrooptic modulators. They could also be used to transmit two independent signals in order to double the transmission capacity. The main applications of PMFs are fiber sensors and telecom. Some specific applications also require PMFs. Among them the Laser MegaJoule (LMJ), under construction at Commissariat à l'Energie Atomique/Centre d'Études Scientifiques et Techniques d'Aquitaine in Le Barp near Bordeaux, France, and that will deliver 1.8 MJ on a few millimeters deuterium–tritium (D–T) capsule in a few nanoseconds employs PMF in the "source," where the laser pulses are shaped temporally and spectrally [3]. The SOP is a key parameter here because it allows taking advantage of the Brewster angle to avoid power loss through reflections in the 240 free-space amplifying chains and isolating the amplifiers with polarizers and Pockel's cells for fast SOP rotation.

Care has been taken to avoid optical power coupling between the orthogonal axes of PMFs with a strong group birefringence. As a matter of fact, polarization-extinction ratios (PER), being the ratio between the optical power of the signal on the SOP on which it is supposed to propagate and the optical power on the orthogonal SOP, as high as 50 dB are usual on the market. However, slight constraints on the fiber dramatically decrease the PER [4], [5]. Usually, these constraints occur at the fiber ends: connectors, splicing, V-grooves, etc. PERs down to 20 dB or even less are common values. It is known that this induces a somewhat low time-varying optical power reduction [6]. Unfortunately, as we shall see, because of the strong group birefringence of PMF, it also induces distortions of any signal as long as its spectrum bandwidth is on the order or larger than the inverse of the differential group delay (DGD) of the PMF [7], [8]. The DGD is the difference in time arrival between both axes of the PMF. Typical DGD values range between 1.5 to 2 ps/m of fiber. Hence, only one long polarization-maintaining (PM) patch cord or a concatenation of a dozen of 1-meter long patch cords is sufficient to distort a 10-GHz-wide signal.

In the following, we will explain in details what happens in a PM patch cord and assess the distortions of pulses traveling in the PMF (Section III). Then, we will extend our numerical and analytical models to a concatenation of fibers (Section IV). Finally, we will show the solutions to avoid distortions (Section V). Among them, we will introduce a new solution [9] (axis alternation) that has been assessed analytically, numerically, and experimentally.

All along this paper, we will apply the models to typical pulses of the LMJ. A phase modulation, i.e., a frequency modulation (FM) is applied. Though the propagation over PMFs, part of this FM is converted into an intensity modulation, i.e., an amplitude modulation (AM). Therefore, we will first introduce the features of the LMJ pulses and its distortion criterion α called the FM-to-AM conversion factor (Section II). However, most results also apply to a telecom signal, typically 10- or 40-Gb/s nonreturn-to-zero, return-to-zero, or differential phase-shift keying signals. As a matter of fact, the spectrum bandwidth of a telecom signal is equivalent to the one of LMJ pulses. Hence, telecom signals are also distorted during propagation over PMFs and bit error rate is degraded. For digital signals, the eye aperture, e.g., Q' [10], should be calculated instead of the FM-to-AM conversion factor α.

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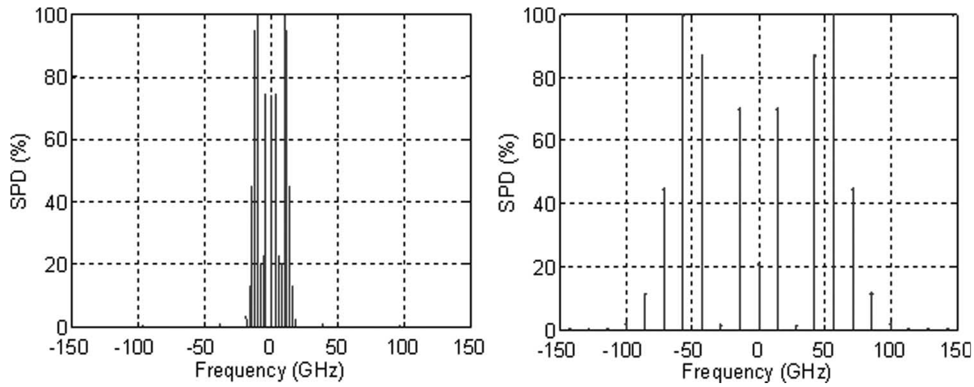

Fig. 1. Spectral power density (SPD) of the phase modulated signal in the case of the anti-Brillouin (left) and of smoothing (right).

II. LMJ Pulse Features

A. Aim of the LMJ

In order to study the fusion of deuterium and tritium both for military and civil applications, it has been decided to build the LMJ, a very high-power laser that delivers 1.8 MJ in a few nanoseconds on a few millimeter capsule of D–T [3]. Such high energies and powers imply to spread the pulse over 40 m^2 in order to avoid damage of the laser itself before reaching the target. These 40 m^2 are divided into 240 40×40 cm^2 beams. The pulses will focus symmetrically and synchronously inside a small oven, called hohlraum where the capsule is placed. The working wavelength for amplification is 1053 nm but it is converted at 351 nm just before focusing on the target in order to increase the photon energy. The total gain of the chain is around 10^{15} and the size of the main building is around $100 \times 300 \times 75$ m^3.

The main requirements on the laser for physical experiments are

1) an almost perfect symmetry of the 240 beams around the hohlraum;
2) a high energy (1.8 MJ);
3) a defined temporal shape; and
4) a homogeneous focal spot.

Overcoming the last requirement is not easy. As a matter of fact, it is impossible to guarantee a perfect spatial phase profile during propagation for the very high-power laser. Hence, when the beam is focused on the target, the focal spot is speckled. This speckle is smoothed by rapid variations of the focal spot that are obtained through wavelength variations and a dispersive element such as a grating [11]. Wavelength variations correspond to a spectral shaping that we now consider.

B. Spectral Shaping of the Pulse

The pulses of the LMJ are shaped temporally and spectrally in the "source." The source uses fiber-based technology similar to what can be found in the telecom industry. Because the SOP is essential in the amplifying chains, PMFs are used. A $\mathrm{LiNbO_3}$ Mach–Zehnder modulator is used to precisely shape the intensity of the pulse. The pulses lasts a few nanoseconds with intensity variation having a typical time scale on the order of a few 100 ps. The induced spectral width is neither sufficient to avoid transverse Brillouin scattering in the chain nor to smooth the beam. Hence, the spectrum is broadened through two sinusoidal phase modulations at 2- and 14.25-GHz frequency f_m with respective modulation index m of 7 and 5 [12]. In the following, we will neglect the impact of the initial intensity modulation.

For simplicity, we will assume that either one or the other phase modulation is applied and not both at the same time. Since, the spectrum bandwidth is mainly determined by the phase modulation, we will not consider the temporal shape of the pulse to determine the spectrum. Hence, the optical field is given by

$$a_0(t) = \exp\left(im \cdot \sin(2\pi f_\mathrm{m} t)\right). \tag{1}$$

We will define its spectrum by a Fourier transform with the following convention: $A(f) = \int a_0(t) \exp[-i2\pi f t]dt$. We also have: $a_0(t) = \int A(f) \exp[i2\pi f t]df$.

Since the signal is periodic, the spectrum is composed of uniformly separated Dirac peaks. It is given by

$$A(f) = \sum_{k=-\infty}^{+\infty} J_k(m) \cdot \delta(f - k \cdot f_\mathrm{m}) \tag{2}$$

where $J_k(m)$ is the kth Bessel function. 98% of the signal energy is included between $\Delta f = 2 \cdot (m+1) \cdot f_\mathrm{m}$ meaning 32 GHz in the case of anti-Brillouin scattering ($f_\mathrm{m} = 2$ GHz, $m = 7$) and 171 GHz in the case of smoothing ($f_\mathrm{m} = 14.25$ GHz, $m = 5$) as can be seen in the spectral power densities shown in Fig. 1. When both phase modulations are applied, the spectrum is the convolution of both and the spectral width is 203 GHz, i.e., 0.75 nm at the central wavelength of 1053 nm.

C. Distortion Criterion

Any linear filtering of the signal that modifies either the relative amplitude or the relative phases of the Dirac peaks distorts the pulses. A problem occurs when part of the phase modulation, i.e., the FM, is converted into intensity modulation, i.e., AM, because the temporal pulse shape may not satisfy the requirement of the fusion experiments. Thus, the distortion

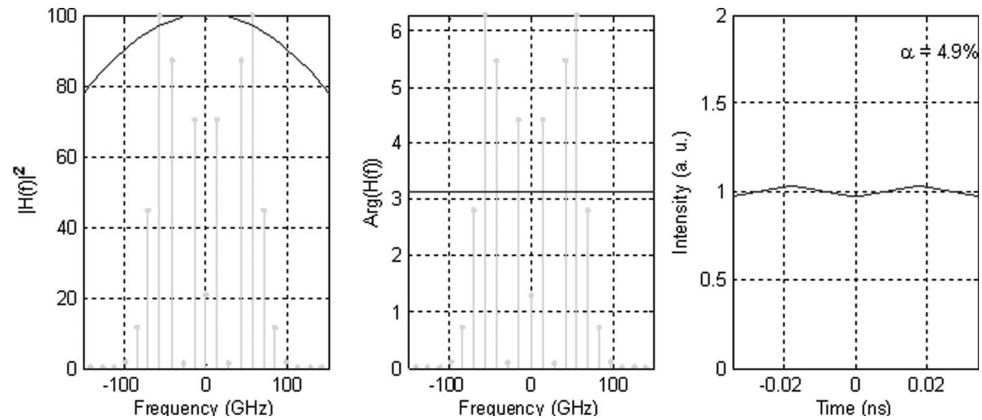

Fig. 2. Filtering function in intensity (left) and in phase (middle), and resulting FM-to-AM conversion (right) of a smoothed signal ($m = 5$ and $f_{\mathrm{m}} = 14$ GHz). $f_{\mathrm{c}} = 0$, $\sigma_f = 10^{-23}\ \mathrm{s}^2$.

criterion α, called the FM-to-AM conversion factor, has been defined by [7], [8]

$$\alpha = 2 \cdot \frac{P_{\max} - P_{\min}}{P_{\max} + P_{\min}} \tag{3}$$

where $P_{\max}$ and $P_{\min}$ are the maximum and minimum optical powers over one phase modulation period $1/f_{\mathrm{m}}$. α ranges from 0% to 200% and is ideally equal to 0 (no intensity variation). It should be limited to a few percent at the output of the LMJ source.

In order to show the impact of a linear transfer function on α, we now consider a function in intensity, e.g., spectral narrowing because of high gain in the amplifiers, and a function in phase, e.g., chromatic dispersion of the fiber in the source. We will show in Section III that the pulse distortion due to the propagation in PM patch cords is also a linear transfer function in intensity and in phase.

D. Intensity Transfer Function

Let us consider the following intensity filtering function

$$H(f) = 1 - \sigma_f (f - f_{\mathrm{c}})^2 \tag{4}$$

f_{c} is the central frequency of the filter and σ_f is related to its width.

The optical field at the output of the filter is equal to

$$a_{\mathrm{out}}(t) \approx \int A(f)\sqrt{H(f)}\exp[i2\pi f t]\, df. \tag{5}$$

If we assume that the filter is wide as compared to the signal spectral width

$$a_{\mathrm{out}}(t) \approx \int A(f)\left[1 - \frac{1}{2}\sigma_f (f - f_{\mathrm{c}})^2\right]\exp[i2\pi f t]\, df. \tag{6}$$

Thus

$$a_{\mathrm{out}}(t) \approx a_0(t) + \frac{\sigma_f}{8\pi^2}\exp\left[i2\pi f_{\mathrm{c}} t\right]\frac{\partial^2}{\partial t^2}\left[\exp\left(-i2\pi f_{\mathrm{c}} t\right) a_0(t)\right]. \tag{7}$$

Replacing (1) in (7) and neglecting the imaginary part (second-order contribution), we obtain

$$a_{\mathrm{out}}(t) \approx \left[1 - \frac{\sigma_f}{2}\left[f_{\mathrm{c}} - m f_{\mathrm{m}}\cos(2\pi f_{\mathrm{m}} t)\right]^2\right] A_0(t). \tag{8}$$

Therefore, the intensity is given by

$$I_{\mathrm{out}}(t) \approx 1 - \sigma_f\left[f_{\mathrm{c}} - m f_{\mathrm{m}}\cos(2\pi f_{\mathrm{m}} t)\right]^2. \tag{9}$$

If f_{c} is lower than $m \cdot f_{\mathrm{m}}$, i.e., the filter is almost centered then the FM-to-AM conversion factor is given by

$$\alpha = \frac{\sigma_f (f_{\mathrm{c}} + m f_{\mathrm{m}})^2}{1 - \frac{1}{2}\sigma_f (f_{\mathrm{c}} + m f_{\mathrm{m}})^2}. \tag{10}$$

If f_{c} is larger than $m \cdot f_{\mathrm{m}}$, α is given by

$$\alpha = \frac{4\sigma_f m f_{\mathrm{m}} f_{\mathrm{c}}}{1 - \sigma_f\left(f_{\mathrm{c}}^2 + m^2 f_{\mathrm{m}}^2\right)}. \tag{11}$$

Note that if the filter is perfectly centered ($f_{\mathrm{c}} = 0$), the AM frequency is $2 \cdot f_{\mathrm{m}}$ while for shifted filtering function the first harmonic is f_{m}.

For smoothing ($m = 5$ and $f_{\mathrm{m}} = 14$ GHz), $f_{\mathrm{c}} = 0$ and $\sigma_f = 10^{-23}\ \mathrm{s}^2$, (10) gives 5% which is what is found numerically (Fig. 2). With $f_{\mathrm{c}} = 100$ GHz and $\sigma_f = 10^{-23}\ \mathrm{s}^2$, (11) gives 33% which is also confirmed numerically (Fig. 3). In the case of anti-Brillouin ($m = 7$, $f_{\mathrm{m}} = 2$ GHz), the spectrum is thinner and the distortion is reduced: $\alpha = 0.2\%$ with $f_{\mathrm{c}} = 0$ and $\alpha = 6\%$ with $f_{\mathrm{c}} = 100$ GHz.

Note that the power variation within the signal spectral bandwidth gives the order of magnitude of α. This confirms results obtained in [7] and [8].

This effect may be compensated for by a filter with the inverse transfer function.

E. Phase Transfer Function

A similar approach may also be done for chromatic dispersion. The transfer function is given by

$$H(f) = \exp\left(-i\pi\frac{\lambda_0^2 D}{c} f^2\right). \tag{12}$$

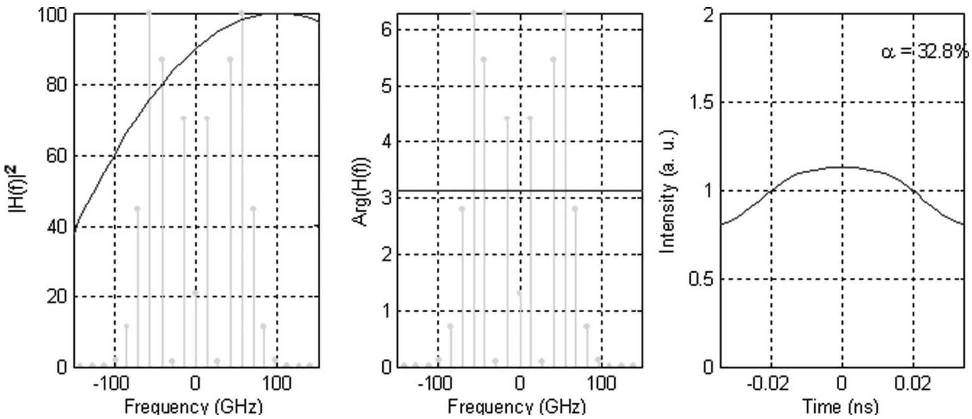

Fig. 3. Filtering function in intensity (left) and in phase (middle) and resulting FM-to-AM conversion (right) of a smoothed signal ($m = 5$ and $f_{\mathrm{m}} = 14$ GHz). $f_{\mathrm{c}} = 100$ GHz, $\sigma_f = 10^{-23}$ s^2.

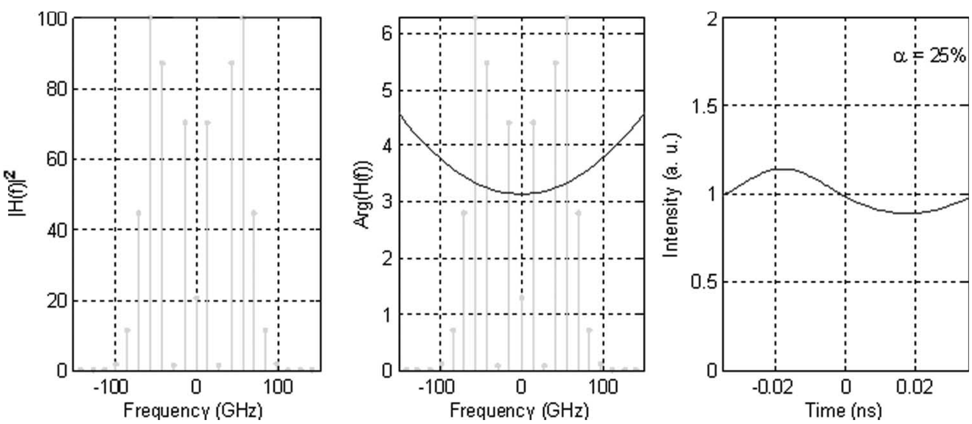

Fig. 4. Filtering function in intensity (left) and in phase (middle) and resulting FM-to-AM conversion (right) of a smoothed signal ($m = 5$ and $f_{\mathrm{m}} = 14$ GHz). $D = -5.5$ ps/nm.

Using a series expansion of (12), one finds

$$\alpha = \frac{4\pi}{c}\lambda^2 |D| m \cdot f_{\mathrm{m}}^2. \tag{13}$$

For 100 m of PM fiber at 1053 nm, $D = -5.5$ ps/nm and, according to (13), $\alpha = 25\%$ for a smoothed pulse. This is confirmed numerically (Fig. 4). In the case of anti-Brillouin $(m = 7, f_{\mathrm{m}} = 2$ GHz$)$, $\alpha = 0.7\%$.

A fiber Bragg grating may be used to compensate for chromatic dispersion.

We now show why propagation over PM patch cords induces linear transfer functions.

III. Propagation Over One PM Patch Cord

A. *Transfer Function of One PM Patch Cord*

PER degradation is due to the fibers themselves, to their use (their bending for instance) but the main contribution comes from the connectors or other constraints at the fiber ends (V-grooves for instance). As a matter of fact, PER of fibers is usually above 50 dB while the PER of connectors is often specified to be only above 20 or 23 dB... and is sometimes below according to our precise measurement using a new scheme [5]. Neglecting the fiber contribution, this means that 1% of the energy (10% in amplitude) of the signal is traveling on the unwanted axis. We name "ghost" the signal replica traveling on the unwanted axis. Because of the PMF DGD, the signal and its ghost have different arrival times at the other end of the fiber.

They will interfere because of the second connector PER degradation (see Fig. 5) [9].

One PM patch cord thus behaves like any two-wave interferometer such as a Michelson or a Mach–Zehnder interferometer, except that the splitting ratio is not 50/50. The interference condition and hence the relative powers on both axes depends on the relative phase of the signal and its ghost. This relative phase is wavelength dependent because it is only a fraction of the optical path difference, namely the DGD. Thus, a PM patch cord, with connectors at both ends, exhibits a frequency dependent PER [9]. If we consider both axes of the fiber together, the distortion of the signal is negligible because, if the energy at a given frequency is not on one axis, it is on the other. However, the aim of PMFs is usually to consider only the fraction of the signal on the axis on which it has been launched. On this axis, the different frequencies of the signal do not have the same transmission and the signal is distorted. We will now compute the transfer function and apply it to an LMJ pulse defined by (1).

The Jones transfer matrix of one PM patch cord is given by

$$M(f) = M_1(f) \cdot R(\theta_0) \tag{14}$$

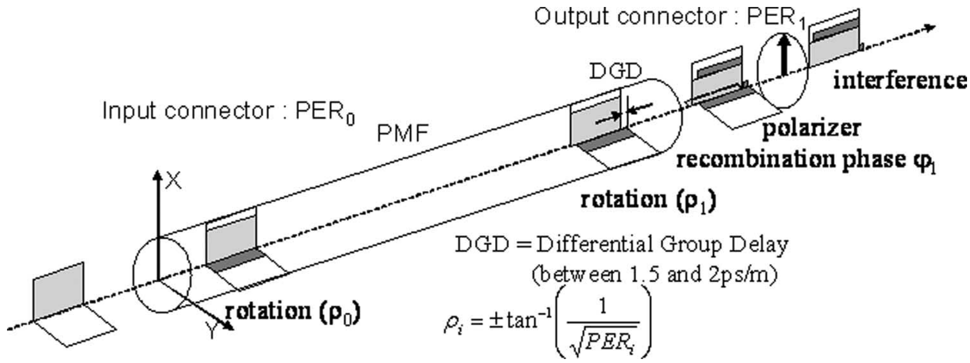

Fig. 5. Pulse progation over one PM patch cord. The input pulse begets a ghost due to rotation (ρ_0) that travels at a different speed. The ghost begets a second ghost due to rotation (ρ_1) that interferes with the main pulse.

where $R(\theta_j = 0)$ is the rotation due to the PER of the first connector PER_0

$$R(\theta_i) = \begin{bmatrix} \cos(\theta_j) & -\sin(\theta_j) \\ \sin(\theta_j) & \cos(\theta_j) \end{bmatrix} \tag{15}$$

and $M_j = 1(f)$ the matrix corresponding to the birefringence of the fiber and the second connector

$$M_j(f) = R(\theta_i) \cdot \begin{bmatrix} 1 & 0 \\ 0 & \exp\left[-i(2\pi\Delta\tau_j f + \varphi_j)\right] \end{bmatrix} \tag{16}$$

where $\Delta\tau_i$ is the PMF DGD and φ_j the relative phase between both axes. As a matter of fact, the DGD corresponds to a time shift $\delta(t - \Delta\tau_j)$ whose Fourier transform is $\exp(-i2\pi\Delta\tau_j f)$. The rotation angle θ_j is determined by PER_j considering the definition of PER [5]

$$\mathrm{PER}_j = \frac{1}{\tan^2(\theta_j)}. \tag{17}$$

The filtering function $H(f)$ is then given by

$$\begin{bmatrix} H(f) \\ 0 \end{bmatrix} = P \cdot M(f) \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} \tag{18}$$

where

$$P = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \tag{19}$$

is the Jones matrix of a linear polarizer. We get

$$H(f) \propto 1 + \frac{1}{\mathrm{PER}_a} \exp\left(i(2\pi f \Delta\tau_1 + \varphi_1)\right) \tag{20}$$

where

$$\mathrm{PER}_a = \sqrt{\mathrm{PER}_0 \cdot \mathrm{PER}_1}. \tag{21}$$

$H(f)$ is plotted in Fig. 6. The frequency scale is given by the inverse of the DGD of the PMF and the intensity and phase variation scale of the filtering function is determined by the PERs of the connectors. The recombination phase φ_1 is a fraction of the DGD. Thus, it is time varying with temperature, pressure, bending... It is almost impossible to control φ_1. Thus, the filtering function will move in frequency along the time and is thus very difficult to compensate for.

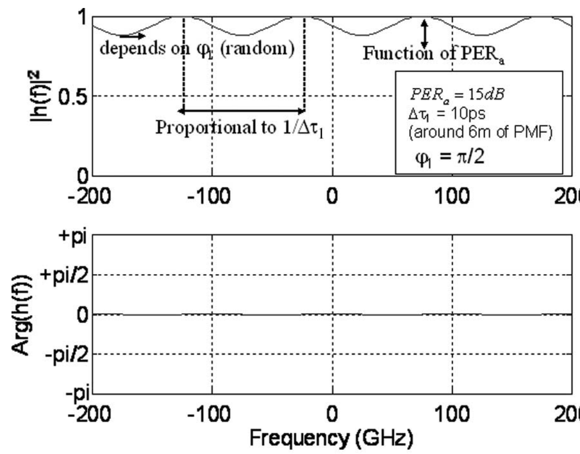

Fig. 6. Transfer function $H(f)$ of one PM patch cord.

B. Impact on a Continuous Wave (CW)

For a CW, the frequency is fixed. However, since $H(f)$ varies with time, the CW will suffer from phase and amplitude variation [6].

C. Impact on an LMJ Pulse

A signal with a relatively broad spectrum (32 GHz in the case of anti-Brillouin and 171 GHz in the case of smoothing) will suffer from variable loss as a CW. However, the variable loss is somewhat averaged by the spectral width because the different frequency components will experience a different loss. The main impact is variable signal distortion. We therefore now compute α due to one PM patch cord.

The inverse Fourier transform of $H(f)$ corresponds to interference between two time-delayed optical waves. Thus, the optical field at the output is

$$a_{\mathrm{out}}(t) = a(t) + \frac{1}{\mathrm{PER}_a} a(t - \Delta\tau_1) \cdot \exp(i\varphi_1). \tag{22}$$

Inserting (1) in (22), one finds the following intensity

$$I_{\mathrm{out}}(t) = 1 + \frac{1}{\mathrm{PER}_a^2} + \frac{2}{\mathrm{PER}_a} \times \cos\left\{2m\sin[\pi f_{\mathrm{m}}\Delta\tau_1]\cos\left[2\pi f_{\mathrm{m}}\left(t - \frac{\Delta\tau_1}{2}\right)\right] - \varphi_1\right\}. \tag{23}$$

TABLE I
EVOLUTION OF α VERSUS FIBER LENGTH FOR RELATIVELY SHORT FIBER LENGTHS IN THE CASE OF ANTI-BRILLOUIN

PMF length	Minimum value of $\varphi_1 = 0\ (2\pi)$	Maximum value of $\varphi_1 = \pi/2\ (\pi)$
$0 < L < 11$ m $\lvert 2m.\sin[\pi f_m.\Delta\tau_1]\rvert \leq \pi/2$	α_{min}	α_{max}
$11\text{m} < L < 22$ m $\pi/2 \leq \lvert 2m.\sin[\pi f_m.\Delta\tau_1]\rvert \leq \pi$	α_{min}	α_{max0}
$L > 22$ m $\lvert 2m.\sin[\pi f_m.\Delta\tau_1]\rvert \geq \pi$	α_{max0}	α_{max0}

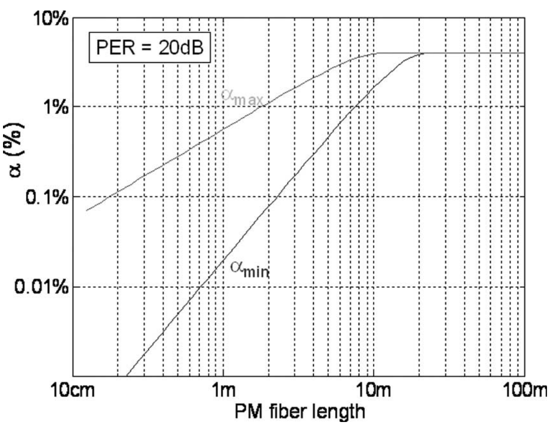

Fig. 7. Variation of the maximum and minimum values of α versus PM fiber length in the case of anti-Brillouin with PER $= 20$ dB.

Thus, α is maximum when

$$2m \cdot \sin[\pi f_{\mathrm{m}} \cdot \Delta\tau_1] \geq \pi. \tag{24}$$

The DGD $\Delta\tau_1$ is given by the fiber length. We measured a DGD of 1.6 ps/m at 1053 nm. For moderate fiber lengths $(f_{\mathrm{m}} \cdot \Delta\tau_1 \ll 1)$, (22) means that α is maximum above 22 m for anti-Brillouin and above 4.5 m for smoothing. The maximum value of α is

$$\alpha_{\max 0} = \frac{4\mathrm{PER}_a}{1+\mathrm{PER}_a^2} \approx \frac{4}{\mathrm{PER}_a}. \tag{25}$$

This means $\alpha_{\max 0} = 4\%$ with $\mathrm{PER}_a = 20$ dB. Increasing the PER reduces the distortions. For shorter lengths, α depends on PER_a, $\Delta\tau_1$, and φ_1. Results are shown in Table I where $\alpha_{\max}$ and $\alpha_{\min}$ are defined by

$$\alpha_{\max} = \alpha_{\max 0} \cdot \left|\sin\left[2m\sin(\pi f_{\mathrm{m}}\Delta\tau_1)\right]\right| \tag{26}$$

$$\alpha_{\min} = \frac{4}{\mathrm{PER}_a} \frac{\sin^2\left[m\sin(\pi f_{\mathrm{m}}\Delta\tau_1)\right]}{1+\frac{1}{\mathrm{PER}_a^2}+\frac{2}{\mathrm{PER}_a}\cos^2[m\sin(\pi f_{\mathrm{m}}\Delta\tau_1)]}. \tag{27}$$

Numerical simulations are in perfect agreement with this analytical computation. Note that the maximum value of α is obtained when $\varphi_1 = \pi/2$, which corresponds to the highest variation of the transfer function within the signal spectrum. For long lengths of fiber, the frequency scale of the transfer function is so low that the different components of the signal spectrum are scrambled and $\alpha = \alpha_{\max 0}$. Fig. 7 shows the variation of α versus fiber length according to Table I. Reducing the fiber length increases the frequency scale of the transfer function and therefore strongly reduces the distortions.

For very long fiber lengths, $f_{\mathrm{m}} \cdot \Delta\tau_1$ is not small. When $f_{\mathrm{m}} \cdot \Delta\tau_1$ is an integer, $I_{\mathrm{out}}(\mathrm{t})$ is constant according to (23) and $\alpha = 0$ whatever φ_1 (see Fig. 8). As a matter of fact, in that case $H(f)$ has a period in the frequency domain equal to f_{m}. All frequency components of the signal will suffer the same intensity and phase variation and the signal is not distorted. This solution to cancel the impact of propagation in PM patch cords cannot be used in practice because these ideal lengths are too high ($f_{\mathrm{m}} \cdot \Delta\tau_1 = 1$ means 312.5 m in the case of anti-Brillouin and 43.8 m for smoothing) and a fiber length both valid for anti-Brillouin and smoothing cannot be found.

To conclude Section III, one should note that short fiber lengths with good PERs are required to avoid distortions. However, it is not always possible to reduce the fiber length and the PERs that we have measured are not sufficiently high [5]. We will show other solutions in Section V but we need first to understand what happens in PM patch-cord concatenation.

IV. PM PATCH-CORD CONCATENATION

A. Transfer Function of a PM Patch-Cord Concatenation

In a concatenation of $1 \ldots j \ldots N$ PM patch cords PER_j, $\Delta\tau_j$, and φ_j may be different from patch cord to patch cord. The filtering function becomes more complicated and random because φ_j is random. It should be studied statistically like polarization-mode dispersion in standard fibers. The statistics of loss for a CW has already been studied [6]. Here, we address the statistics of distortions. Hence, the frequency dependence of $H(f)$ has to be taken into account. Note that the frequency scale will be determined by the length of the fibers and the amplitude scale by the PERs.

The Jones transfer matrix $M(f)$ should be replaced by

$$M(f) = \left[\prod_{j=1}^{N} M_j(f)\right] \cdot R(\theta_0). \tag{28}$$

In order to understand what happens to the signal, we will expand (18) in series considering that PER_j are sufficient high so that θ_j is small. This assumption is reasonable since according to (17), $\mathrm{PER}_j = 20$ dB, means $\theta_j = 5.7°$.

Equations (15) and (16) become

$$R(\theta_i) = \begin{bmatrix} 1 & -\theta_j \\ \theta_j & 1 \end{bmatrix} \tag{29}$$

$$M_j(f) = \begin{bmatrix} 1 & -\theta_j \cdot \exp\lfloor -i(2\pi\Delta\tau_j f + \varphi_j)\rfloor \\ \theta_j & \exp\left[-i(2\pi\Delta\tau_j f + \varphi_j)\right] \end{bmatrix}. \tag{30}$$

If we consider a concatenation of three PM fibers, one gets

$$\begin{aligned} H(f) \approx 1 &- \theta_0\theta_1 e^{-i(2\pi f\Delta\tau_1+\varphi_1)} \\ &- \theta_1\theta_2 e^{-i(2\pi f\Delta\tau_2+\varphi_2)} - \theta_2\theta_3 e^{-i(2\pi f\Delta\tau_3+\varphi_3)} \\ &- \theta_0\theta_2 e^{-i(2\pi f\Delta\tau_1+\varphi_1+2\pi f\Delta\tau_2+\varphi_2)} \\ &- \theta_1\theta_3 e^{-i(2\pi f\Delta\tau_2+\varphi_2+2\pi f\Delta\tau_3+\varphi_3)} \\ &- \theta_0\theta_3 e^{-i(2\pi f\Delta\tau_1+\varphi_1+2\pi f\Delta\tau_2+\varphi_2+2\pi f\Delta\tau_3+\varphi_3)} \\ &- \theta_0\theta_1\theta_2\theta_3 e^{-i(2\pi f\Delta\tau_1+\varphi_1+2\pi f\Delta\tau_3+\varphi_3)}. \end{aligned} \tag{31}$$

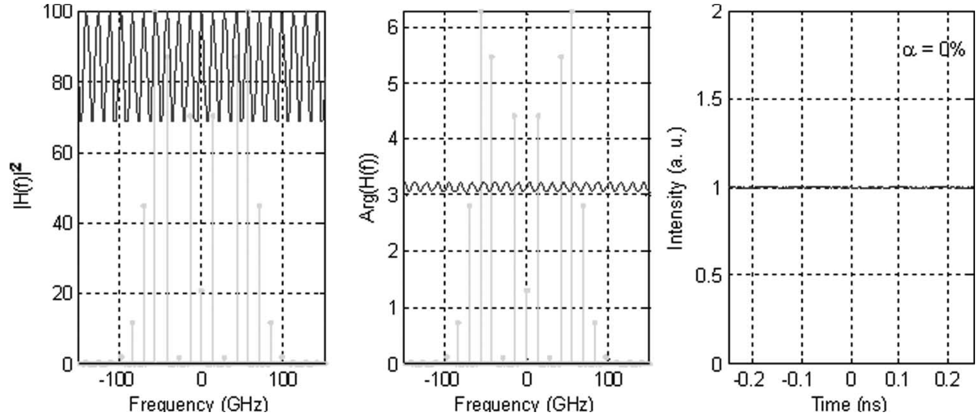

Fig. 8. Filtering function in intensity (left) and in phase (middle) and resulting FM-to-AM conversion (right) of a smoothed signal ($m = 5$ and $f_{\mathrm{m}} = 14$ GHz). The filtering function has the same period in the frequency domain as the signal and thus does not distort the signal.

In the right-hand side of (31), "1" corresponds to the initial pulse. The second-order terms correspond to ghosts of the pulse that are rotated somewhere in the concatenation, that travel along the orthogonal axis, that are rotated back to the axis of the initial pulse and interfere with it. Thus, only the sums of adjacent patch cord DGDs appear to the second order. The fourth-order term in (31) corresponds to a ghost of the initial pulse that travels along the wrong axis in the first patch cord, the right axis in the second, and on the wrong axis in the last section. Thus, it is rotated four times and is somewhat negligible. No pulse with an odd number of rotations will interfere at the end of the concatenation with the principal pulse. Hence, only even orders in θ_i appear in (31). Moreover, up to the second order in θ_i, only the sums of DGDs of adjacent patch cords are relevant. This important feature will be used when presenting the solutions to signal distortion due to propagation in PM patch cords.

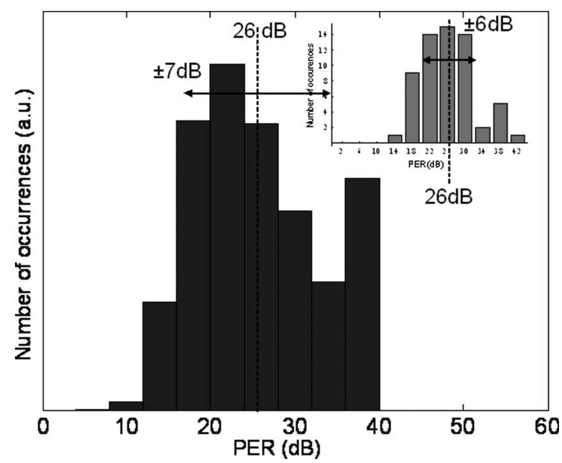

Fig. 9. Histogram of PER in decibels for usual connectors. The rotation angle distribution is supposed to be Gaussian with a standard deviation of 5°. The inset shows the histogram of measurement of 61 connections.

B. Impact on an LMJ Pulse

We have numerically simulated the impact of a concatenation of patch cords on the FM-to-AM conversion factor. We have considered fiber lengths uniformly distributed from 1 to 2 m. The rotation angles θ_i were supposed to follow a Gaussian distribution with a standard deviation of 5°. According to (17), this yields the PER distribution shown in Fig. 9. Since measuring PERs above 40 dB is almost impossible we have replaced PER values higher than 40 by 40 dB. The distribution shown in Fig. 9 is consistent with our 61 measurements of usual connectors that can be found on the market [5]. The mean (26 dB) and the standard deviation (6 dB) is very close to the one (26 and 7 dB, respectively) obtained with our Gaussian assumption of the rotation angle.

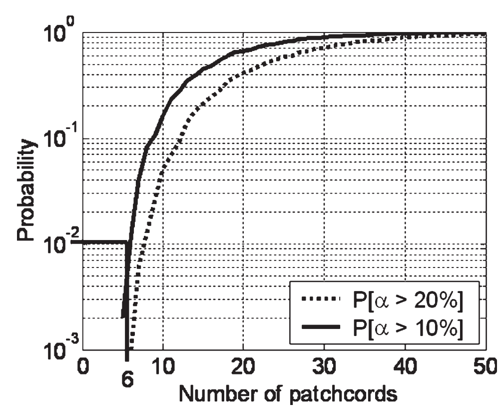

Fig. 10. Probability for α to be above a given value (10% and 20%) versus the number of patch cords in the case of anti-Brillouin (reference case).

The relative phases φ_j were uniformly distributed between 0 and 2π. 1000 random concatenations of 50 patch cords were drawn. The output of our numerical simulations is the distribution of the FM–AM conversion factor α.

Fig. 10 shows the probability for α to be above given values (here 10% and 20%) in the case of anti-Brillouin. Note that high values of α may be reached although the mean value of PER is relatively high (26 dB) because a single poor connection contributes many times to the transfer function according to (31). If one wants to avoid α being above 10% more than 1% of the time, no more than six patch cords can be concatenated with anti-Brillouin. With smoothing, the number is even reduced to 3 since the signal spectrum bandwidth is larger. Hence, it is required to find solutions to overcome these limits. As a matter of fact, only one single-stage optical amplifier may contain ten

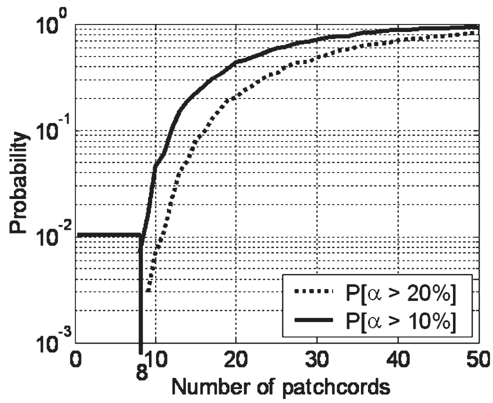

Fig. 11. Probability for α to be above a given value (10% and 20%) versus the number of patch cords in the case of anti-Brillouin with shorter patch cords.

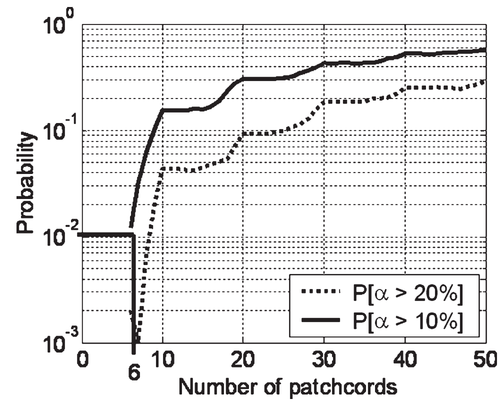

Fig. 12. Probability for α to be above a given value (10% and 20%) versus the number of patch cords in the case of anti-Brillouin with polarizers every ten patch cords.

fibers, not always with connectors but also with constraints at the fiber ends (protected splices for instance). This is the purpose of the next section.

V. Solutions

In the following, we will propose and assess different solutions. The results will be given for the case of anti-Brillouin but it also applies to smoothing.

The simplest solution is to remove the PMF! The alternative is either to use standard single-mode fiber or polarizing fibers. Standard fibers require several active polarization controls because some components are polarization sensitive. Active polarization control is still a challenge today, especially when polarization controllers have to be cascaded. Polarizing fiber seems the ideal solution because it dramatically attenuates the ghost of the signal traveling on the wrong axis. However, polarizing fibers at 1.053 μm are very difficult to manufacture and expensive. As a matter of fact the cutoff wavelength should be different on both axes [13], [14]. Moreover, splicing polarizing fibers is difficult. Therefore, we need to find alternative solutions. These solutions will be considered separately in a first step and then together.

A. Reduction of the Fiber Length

As we have seen in Section III, reducing the fiber lengths will reduce the DGDs and thus increase the frequency scale of the transfer function reducing the induced distortions. In practice, it is difficult to reduce the fiber length below 50 cm because margin is required for splicing. We have simulated the impact of a fiber length reduction by considering a uniform distribution between 50 cm and 1 m (Fig. 11). Up to eight fiber patch cords may be concatenated with this scheme. The improvement is not sufficient.

B. Polarizer Insertion

Adding a polarizer in the fiber concatenation should wipe out the ghost traveling on the wrong axis although it cannot cancel the accumulated FM-to-AM conversion. As we have seen with (31), up to the second order only the sums of DGDs of adjacent fibers have an influence on FM-to-AM conversion. A polarizer removes all the sums composed of fibers before and after the polarizer and thus reduce FM-to-AM conversion of the concatenation. It resets the interferometric state of the PM fiber. We have simulated the impact of adding a polarizer every ten fiber patch cords (Fig. 12).

Results of course do not change for the first nine patch cords but the slope of the curve suddenly changes after each polarizer. Hence, polarizers alone are not sufficient except if added after every two or three patch cords. However, polarizers are interesting when combined with other solutions that already allow bridging more than ten patch cords.

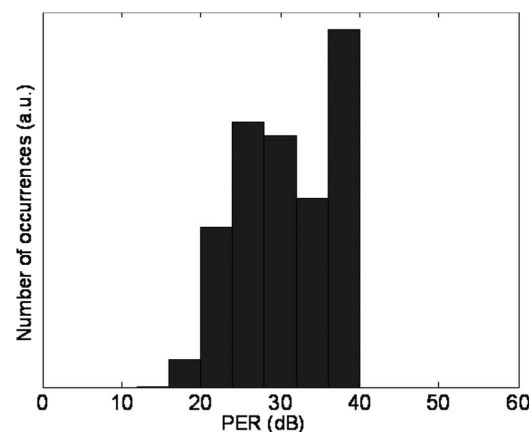

Fig. 13. Histogram of PER in decibels for improved connectors. The rotation angle distribution is supposed to be Gaussian with a standard deviation of 2.5°.

C. Improved Connectors

Some singular vendors are able to mount connectors with far better performance than usual. According to our measurements, we have considered a Gaussian distribution of the rotation angles with a pessimistic standard deviation of 2.5° instead of 5° although we did not have sufficient experimental data to make a fit. This angle distribution corresponds to a PER distribution represented in Fig. 13.

With these connectors, ten patch cords may be concatenated instead of six (Fig. 14).

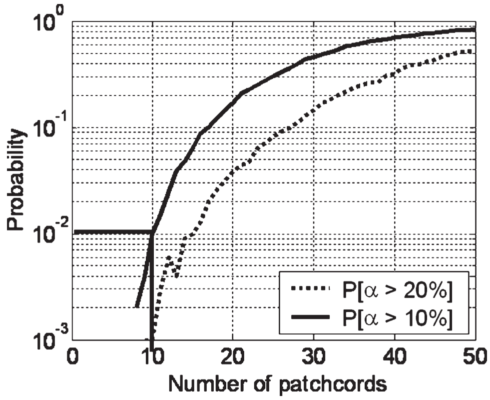

Fig. 14. Probability for α to be above a given value (10% and 20%) versus the number of patch cords in the case of anti-Brillouin with improved connectors.

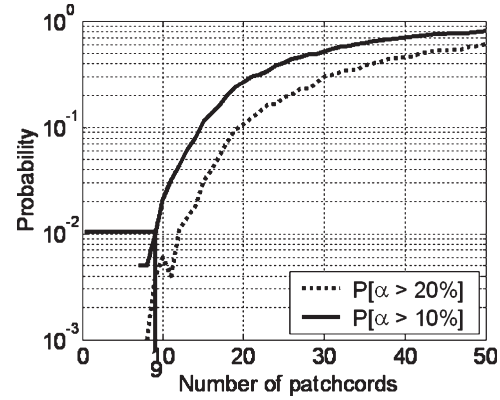

Fig. 16. Probability for α to be above a given value (10% and 20%) versus the number of patch cords in the case of anti-Brillouin with axis randomizing.

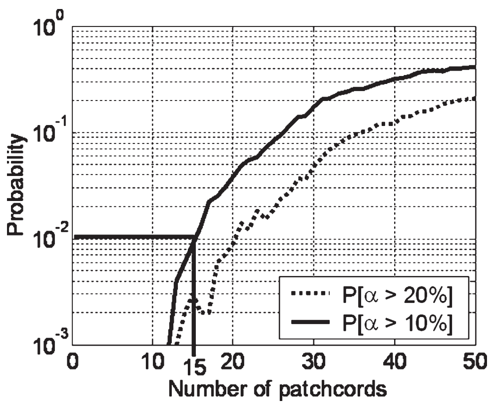

Fig. 15. Probability for α to be above a given value (10% and 20%) versus the number of patch cords in the case of anti-Brillouin with axis alternation.

D. Axis Alternation

In the following, we will show a new solution at no cost [9]: the DGD of a fiber concatenation may be canceled by adding another PM fiber having the appropriate length and inverting the fast and the slow axes. In practice, there is no need to add patch cords: the solution just consists in alternatively aligning the connector key with the slow and the fast axes. Hence, the DGD of one patch cord is canceled by the following one. This idea has been checked experimentally [9].

Alternating the PMF axes was already known in a different context: it usually consisted of canceling the impact of DGD when the input SOP was unknown while PM fibers had to be used to make two signals interact. This was the case for some filtering functions or for time demultiplexing based on Kerr effect for instance [15], [16]. In our case, alternating the axes was not considered up to now because the signal was supposed to travel on a single axis of the PMF while it is not because of the somewhat low PERs at fiber ends.

We have simulated the axis alternation on FM-to-AM conversion (Fig. 15) assuming that the PER distribution is the same as the one of our reference (Fig. 9). It is not perfectly canceled because of the PERs of connectors between one patch cord and the following one. Moreover, we did not pair the lengths of the patch cords: the fiber length distribution is still uniform between 1 and 2 m. Even though, up to 15 patch cords may be concatenated instead of six without axis alternation. If care is taken to the length (all the patch cords have a length of 1.5 m instead of ranging from 1 to 2 m), the number of patch cords is extended to 17.

Note that axis alternation is required: canceling the total DGD at the end would not be efficient. As a matter of fact, we have seen from (31), that only the sums of DGDs of adjacent patch cords have an influence on the signal distortion. The DGD of a patch cord with reversed axes should be considered as negative. Assuming that all the patch cords have exactly the same length, all the sums with an even number of terms are equal to zero while all the remaining sums are minimal and equal to the DGD of a single patch cord. Hence, distortions are minimized. To assess this feature, we have also simulated a concatenation of fibers having a random orientation of axes. On average, for a long concatenation, the total DGD is also canceled but the sums of DGDs of consecutive patch cords are not as much reduced. Results are given in Fig. 16. No more than nine patch cords may be concatenated instead of 15 with axis alternation and six with constant axis orientation.

E. Cumulated Solutions

Note that axis alternation is almost a sufficient solution if one is able to fuse two fibers with axis alternation and low constraints as shown in [17]. As a matter of fact, any patch cord could just be cut in two and refused with axis alternation. Nevertheless, because of the poor performance of usual connectors, it may happen that the signal rotates a lot and that most of the optical power is not on the axis on which the signal is supposed to be. Hence, if a long concatenation of PM fibers is used, high-performing connectors are preferable.

If we now combine all the solutions considered in Sections V-A–D (short patch cords with improved connectors, axis alternation from patch cord to patch cord and a polarizer every ten patch cords) we almost cancel FM-to-AM conversion even for 50 patch cords (Fig. 17). Note that in Fig. 17, the scale has been changed and the distortion thresholds are now 1% and 2% instead of 10% and 20%. The ultimate limit in terms of patch cords will now come from power fluctuations [6] that may be reduced with optical amplifiers working in saturation regime.

This solution is also valid in the case of smoothing as shown in Fig. 18. Here, also the scale is 1% and 2%.

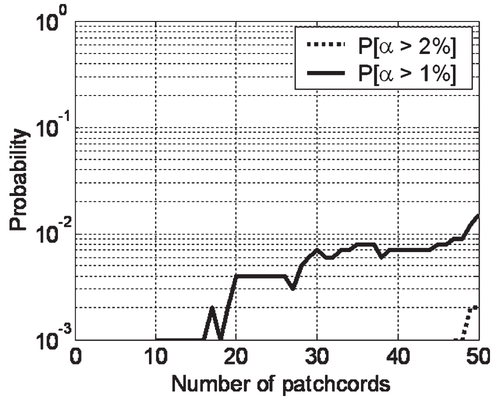

Fig. 17. Probability for α to be above a given value (1% and 2%) versus the number of patch cords in the case of anti-Brillouin with short patch cords with improved connectors, axis alternation from patch cord to patch cord and a polarizer every ten patch cords Note that the scale of distortion has been changed to 1% and 2% instead of 10% and 20%.

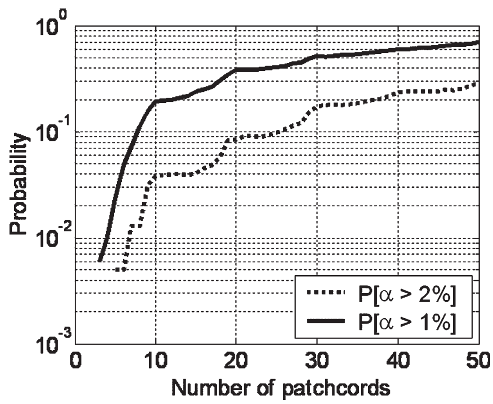

Fig. 18. Same as Fig. 17 but in the case of smoothing.

F. Solution With Length Constraints

In practice, it is not always possible to reduce the length of a fiber to the minimum, i.e., between 50 cm and 1 m. For instance, in an optical amplifier, the doped fiber length is determined by the amplifier features (gain, noise, saturation power). We therefore now consider the case of a 6-m-long doped fiber between two polarizing isolators. The question is to find the best pigtail length for the isolators, assuming that axes are crossed from the pigtails to the doped fiber. On one hand, very short isolator pigtails reduce their contribution to the distortions of the signal. On the other hand, longer pigtails may compensate for part of the impact of the doped fiber. We have simulated the distortions of such a concatenation of three fibers (Fig. 19).

A minimum is obtained when the length of the side fibers (the isolator pigtails) is half the length of the central (doped) fiber. We are able to explain this phenomenon with (31) by calculating the sums of the DGDs of adjacent fibers. We have normalized the DGD of the central fiber to 1. Results are shown in Table II.

If the pigtails have the same length as the doped fiber ($\Delta\tau_1 = \Delta\tau_2 = \Delta\tau_3 = 1$), among the 6 sums of the DGDs of adjacent fibers, 2 sums are equal to 0 and 4 sums to 1. This is the same when the pigtails are very short ($\Delta\tau_1 = \Delta\tau_3 = \varepsilon \ll \Delta\tau_2 = 1$). Hence, this explains why the probabilities for α to reach 0.5% or 1% are almost the same for 0.5 and 6 m in Fig. 19. However, if the pigtail length is half the one of the doped fiber, only one sum is equal to 1 and 4 sums are equal to 0.5 increasing the frequency scale by a factor of 2. Hence, this case is favorable to reduce distortion.

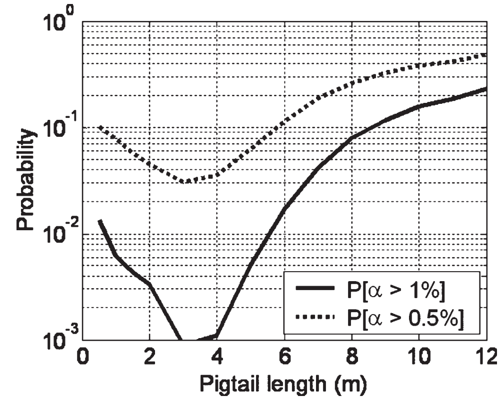

Fig. 19. Probability for α to be above a given value (0.5% and 1%) versus the length of the pigtail in the case of a doped fiber connected by two isolators with pigtails.

TABLE II
SUMS OF THE DGDs OF ADJACENT FIBERS IN THE CASE OF THREE CONCATENATED FIBERS. THE DGD OF THE CENTRAL FIBER, $\Delta\tau_2$, IS SUPPOSED TO BE CONSTANT AND EQUAL TO 1

	-1 1 -1	-½ 1 -½	-ε 1 -ε
$\lvert\Delta\tau_1\rvert$	1	½	0
$\lvert\Delta\tau_2\rvert$	1	1	1
$\lvert\Delta\tau_3\rvert$	1	½	0
$\lvert\Delta\tau_1+\Delta\tau_2\rvert$	0	½	1
$\lvert\Delta\tau_2+\Delta\tau_3\rvert$	0	½	1
$\lvert\Delta\tau_1+\Delta\tau_2+\Delta\tau_3\rvert$	1	0	1
Summary	4 x 1	4 x ½ + 1 x 1	4 x 1

G. Experimental Validation

We have applied the solutions presented here to a source prototype of the LMJ including fiber amplifiers [12]. This includes around 30 fiber pigtails and patch cords. It is impossible to sufficiently wait to cover the whole statistics of α and therefore to have perfect comparison between the source with and without the implemented solutions. Nevertheless, α has been measured in the case of anti-Brillouin alone with a measurement bandwidth of 6 GHz. Before using these schemes (except that polarizing components such as modulators were already included), typical measured α values ranged from 5% to 30% with a mean of 15% and a standard deviation of 7%. For the new prototypes, typical values are below a few percents (from 0.5% to 3% with a mean of 1.5% and a standard deviation of 0.8%). The improvement is sufficiently high to make sure that our solutions are relevant.

VI. CONCLUSION

The impact of the finite PER of PMFs on power fluctuations was already studied in the 1980s [6]. A new technique has been developed to accurately measure the polarization extinction ratio of connections [5]. It was found that often 1% or even more of any signal optical power is traveling on the wrong axis. We have shown that it not only induces power fluctuations but also distortions of signal as long as its bandwidth is on the order or larger than the DGDs of the PMF. The signal shown here as an example is a typical source pulse of the LMJ but distortions also apply to any telecom signal since their spectral bandwidths are on the same order. Guided by some analytical modeling, we have compared different solutions to this problem with numerical simulations. The most simple and efficient of them is to alternatively reverse the axes of the PM fibers in order to compensate the DGD of one fiber by the following one [9]. However, all the solutions may also be combined and distortions become negligible. We have successfully applied them to a source prototype of the LMJ.

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Nicolas Beck, photograph and biography not available at the time of publication.

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